

WHITE BOOK

XQE122

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1 Introduction

Suitable for indoor warehousing scenarios—such as rack warehouses and stacked storage facilities—including high-level bin stacking, pallet racking, automated vertical warehouse interfacing, and buffer zone sorting. The vehicle can autonomously navigate, automatically unload with a forklift function, handle automatic loading and unloading from racks, perform targeted rack stacking, and avoid obstacles automatically. A single unit meets diverse needs such as inbound and outbound operations, nighttime pre-sorting, line-side delivery, and vertical warehouse integration, making it the breakthrough choice for intensive and digitalized warehouse upgrades.

1.1 Intended to use

The XQE122 excels in automated high-level stacking with conventional aisle widths, reaching lifting height of 5.5m to handle pallets, stacked boxes, custom frames and irregular carriers. The compact size, with its length to fork faces (L2) of 1365mm, and the reach-type fork mechanism maximize storage density and enhance space utilization.

It enables distributed dense storage configurations and seamless collaboration with industrial automation equipment (robotic arms) and larger automated systems (WMS system, MES system, ERP system, etc.), positioning it as a transformative tool for modern logistics operations demanding space efficiency and operational continuity.

The system is designed in compliance with ISO 3691-4, ensuring safe operation in industrial settings.

The AGV is intended exclusively for industrial material handling applications and must not be used for personnel transport.



Fig 1.1: XQE122

Operating Modes

1. Automatic Mode (No Rider Allowed)

- Fully autonomous AGV operation
- Personnel transport is strictly prohibited
- All safety systems, including personnel detection and obstacle avoidance, are active
- Intended for automated production operation, continuous material transport, and fleet coordination

2. Maintenance Mode

- Carry out maintenance and operation using a handheld Wired control.



- Personnel detection systems are safely deactivated
- Used in controlled environments under direct operator supervision
- Intended exclusively for service, diagnostics, and maintenance activities
- Vehicle motion and safety functions are limited or disabled as required for maintenance
- Access restricted to trained and authorized personnel only

1.2 Key Functional Overview

Navigation & Safety

- 3D laser SLAM navigation, with the option to be equipped with ground QR codes for auxiliary positioning Suitable for complex and diverse scenarios such as production workshops and large warehouses.
- 360° bottom safety LiDAR, fork-tip Safety bumper and safety PLC monitoring for collision prevention
- Top 3D LiDAR, fork bottom 3D LiDAR and fork tip photoelectricity for obstacle detection
- Emergency stop switches

Material Handling

- Automatic picking and recognition of pallets and material cages.
- Accurate load picking and unloading even with slight pallet misalignment.
- Safety identification for checking whether there is any goods at the unloading storage location
- Real-time storage location status (empty/full) for optimized scheduling.

Operational Flexibility

- Support point-to-point, column-to-column, and collaborative tasks
- Applicable to standard shelves and deep aisle warehouse scenarios
- Manual or automatic charging mode

System Integration

- Communicates with WMS (Warehouse Management System), MES (Manufacturing Execution System), ERP (Enterprise Resource Planning), and other enterprise systems.
- Intuitive touchscreen or PDA, mobile phone and call box controls for easy task assignment and status monitoring.
- Provides real-time operational status, alarms, and task reporting

1.3 Primary Capabilities

The system provides autonomous storage management with automatic detection of storage availability, optimal unloading allocation, and precise pallet handling. It supports collaborative fleet operations, enabling multiple AGVs to perform both independent and synchronized transport tasks.

Task capabilities include standard operations such as point-to-point and column-based movements with intelligent task selection, as well as collaborative functions like docking, action, position coordination, and multi-vehicle binding.

Overall, the system enables flexible, efficient, and scalable multi-AGV operation for complex material handling environments.



1.4 Applications

XQE122 is designed for large-scale, high-throughput industrial environments, supporting long-distance transport, dense pallet storage, heavy-load handling, and continuous operation with high precision and reliability. It is well suited for warehouses and production facilities requiring efficient material flow and integration with automated infrastructure such as elevators, conveyors, doors, palletizers, and robotic arms.

The system features a warehouse-free storage location management approach, enabling autonomous perception, decision-making, and automatic forking without fixed storage definitions. This improves operational efficiency while addressing labor shortages, high turnover, unclear storage zoning, and inefficient stacking or sorting practices.

At the warehouse-area level, XQE122 provides intelligent area-based control, supporting flexible unloading and picking strategies, bidirectional traffic, real-time empty/full status monitoring, and selective or full-area clearing and filling. These functions enhance storage flexibility, space utilization, and unloading efficiency.

Implementation functions such as route teaching, visualized and editable layouts, customizable storage areas, and task definitions enable rapid deployment and site-specific configuration, ensuring adaptability across different operational scenarios.

1.5 Compliance and Safety Standards

The XQE122 is designed and developed in accordance with relevant international and European standards to ensure safe operation, reliable performance, and compliance with applicable regulatory requirements for automated material handling systems.

- **Machinery Directive 2006/42/EC**

The XQE122 complies with the essential health and safety requirements of the European Machinery Directive 2006/42/EC, forming the basis for CE conformity and safe integration into industrial environments.

- **ISO 3691-4**

Industrial trucks — Safety requirements and verification — Part 4: Driverless industrial trucks and their systems

Referenced under the Safety System section:

“The safety system mainly consists of a safety PLC controller and various safety sensors to form a safety monitoring system to meet the requirements of ISO 3691-4 standard.”

- **EMC Directive 2014/30/EU (Electromagnetic Compatibility)**

The system fulfills the requirements of the EMC Directive 2014/30/EU, ensuring that the XQE122 does not generate electromagnetic disturbances that could affect other equipment and is itself resistant to external electromagnetic influences, guaranteeing reliable operation in industrial environments.



1.6 Truck parameters

Table 1.1: Basic Parameters

Parameter	Symbol	Unit	Value
Manufacturer			EP
Model name			XQE122
Model Designation			Indoor Reach Stacking
Drive			Electric
Load Capacity		kg	1200
Service Weight		kg	2880
Navigation			3D slam/QR code
Communication			Wi-Fi/5G
Positioning Accuracy		mm	± 10
Indoor/Outdoor			Indoor

Battery Parameters

Parameter	Symbol	Unit	Value
Battery Type			Li-ion Battery
Battery Voltage/ Nominal Capacity		V/Ah	48/150
Usage Time		Hours	5-6

Dimensions

Parameter	Symbol	Unit	Value
Load Centre Distance	c	mm	600
Load Length	x	mm	318
Wheelbase	y	mm	1265
Forward distance	l4	mm	590
Lifting Height	h3	mm	5500

Other Parameters

Parameter	Symbol	Unit	Value
Travel speed		m/s	1.2 m/s
Max. Gradeability		%	3/5
Maximum Floor Level Deviation		mm	≤20
Turning Radius	Wa	mm	1544

Aisle Requirements

Parameter	Symbol	Unit	Value
Aisle width for pallets 1000×1200	Ast	mm	2450
Aisle width for pallets 800×1200	Ast	mm	1840
Aisle width for one-sided	Ast	mm	3150

Auto-Charge

Parameter	Symbol	Unit	Value
Communication			Wi-Fi
Charging Power			48V/100A



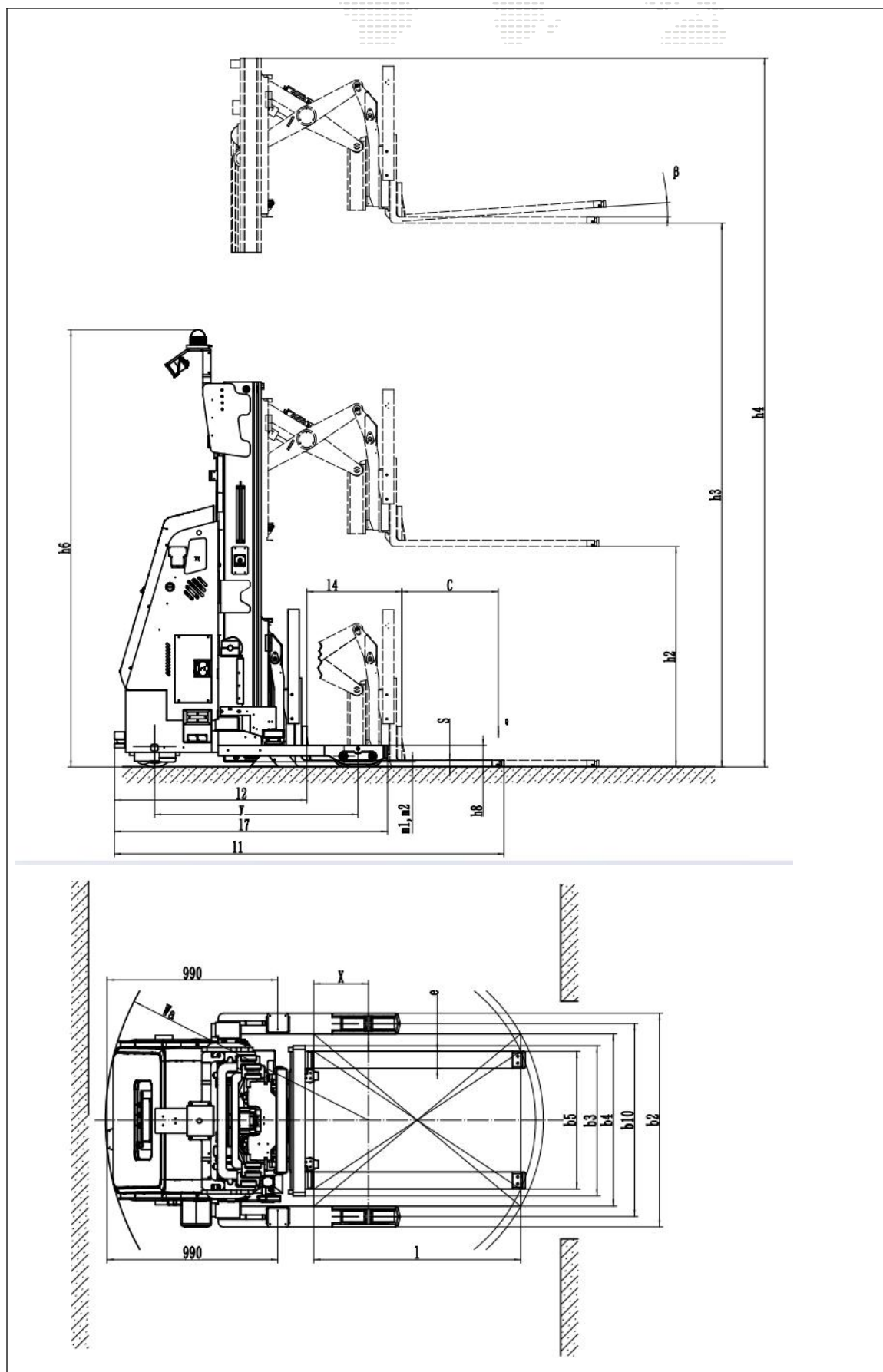


Fig 1.2: Dimensions



1.7 Optional Accessories

1.7.1 Loaders and automatic charging stations

Optional items	Standard specification	Optional specifications
Fork size	100mm*40mm*1200mm	customizable
Automatic charging station	/	48V100AH
Manual charger	48V50A	/

1.7.2 Height of the gantry

Optional items	Type of gantry	lift height	The height of the gantry when it is lowering		Freedom to ascend to greater heights	maximum height	lifting speed	Navigation laser height
			With navigation	Without navigation				
		mm	mm	mm	mm	mm	m/s	mm
Standard specification	3 - Standard Boom Frame	4500	2718	2393	1368	5523	0.17	2618
Optional specifications	2 - Standard Boom Frame	3200	2965	2640	/	4240	0.17	2865
	2 - Standard Boom Frame	3200	2823	2498	/	4173	0.17	2723
	3 - Standard Boom Frame	5500	3053	2728	1700	6825	0.17	2953

1.8 Key Benefits

- Compact indoor reach stacking AMR for high-level stacking (5.5m lifting height) and transport in narrow aisles.
- Perfect for distributed dense storage for pallets, boxes and irregular loads.
- Seamless integration with industrial automation equipment and WMS/WES/ERP system for automated workflows.
- SLAM navigation precision with 10mm positioning accuracy and 3D vision-guided load alignment.
- Intuitive touchscreen or PDA, mobile phone and call box controls for easy task assignment and status monitoring.

2 Main truck components and description

2.1 Physical Layout

These diagrams illustrate the physical layout of major components including navigation sensors, safety devices, control units, drive systems, and material handling mechanisms, providing a clear reference for installation, maintenance, and troubleshooting.





Fig 2.1: Right side Components



Fig 2.2: Left Side Components

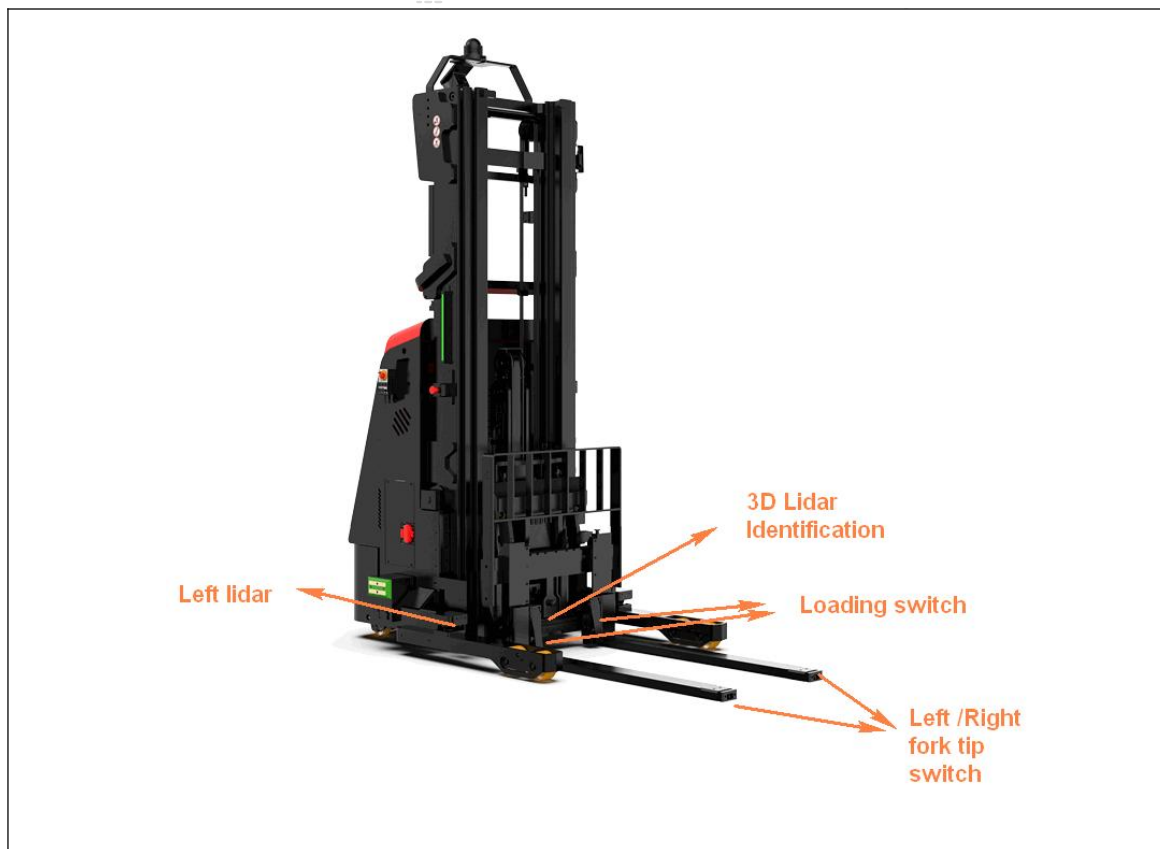
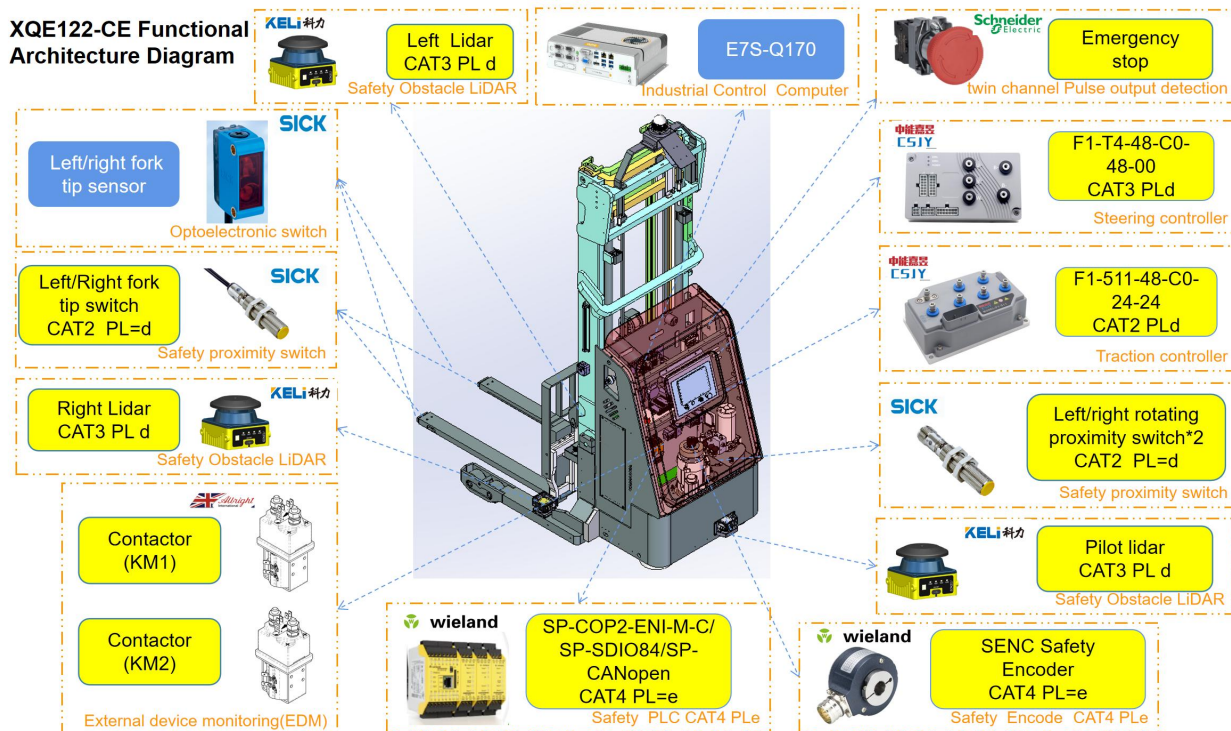


Fig 2.3: Back/Fork side components

2.2 Functional Architecture Diagram



2.3 Navigation System

Laser SLAM (Simultaneous Localization and Mapping) technology enables the generation of precise 3D digital representations of environments. By utilizing LiDAR sensors, the system continuously scans its surroundings while simultaneously determining its own position within the mapped space. The result is a highly detailed 3D point cloud that captures spatial structures such as walls, racks, and obstacles.

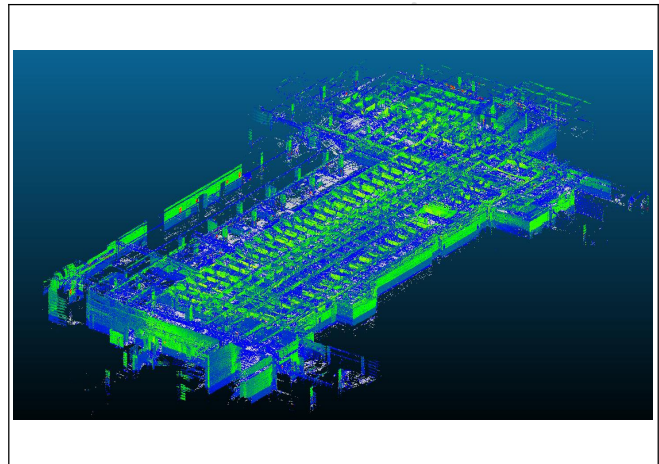


Fig 2.4: 3D-point cloud picture

As illustrated in Fig. 2.4, the generated point cloud provides a comprehensive digital representation of the warehouse environment, allowing the system to perceive depth, geometry, and navigable space with high accuracy. This digital map forms the foundation for reliable and autonomous navigation.

2.4 Navigation Methods

Laser SLAM Guidance

- Purpose: Enables autonomous positioning and route planning in natural environments.
- Working Principle: 3D LiDAR scans the surroundings to build a real-time map; SLAM algorithms compute the vehicle's position relative to the map.
- Benefits: Accurate navigation in large, complex warehouses without physical .

2.5 Blind-spot laser Lidar

- Purpose: Supports obstacle detection by identifying frontal and low-hanging objects, assisting the vehicle in navigating through dynamic environments.
- Working Principle: The 3D laser obstacle avoidance function refers to an intelligent feature that utilizes 3D laser radar (LIDAR) to scan the surrounding space environment in real time, sensing the three-dimensional positions and shapes of obstacles, thereby achieving dynamic obstacle avoidance. 3D laser can obtain height information and can identify suspended obstacles, low obstacles, and complex terrains, such as protrusions at the bottom of shelves, hanging objects, slopes, etc., enhancing the comprehensiveness and safety of obstacle avoidance. The system builds a real-time point cloud model and combines the obstacle avoidance algorithm to dynamically adjust the travel path while ensuring the target route, avoiding collisions.
- Benefits: Enhances operational awareness by assisting in the detection of pedestrians, forklifts, and suspended obstacles. This contributes to smoother navigation and reduces the likelihood of disruptions such as minor collisions or traffic jams in busy warehouse environments.



- Implementation: The overhead-mounted 3D laser radar (LIDAR) continuously scans the ground ahead and nearby obstacles, improving overall environmental perception in real time.

2.6 Bottom 360° Safety LiDAR

- Purpose: Provides comprehensive ground-level coverage to detect surrounding obstacles and support safe vehicle operation.
- Working Principle: The system utilizes three LiDAR sensors (Installed at the front of the vehicle and on the supporting legs) to achieve 360° environmental scanning. The detected data is processed in real time to generate a safety field around the vehicle, which is divided into four configurable zones. In the outer zones, the system issues warnings and progressively reduces vehicle speed as obstacles are detected closer to the vehicle. The innermost zone (red zone) represents the final safety boundary. When an object is detected within this zone, the system triggers an immediate and controlled stop of the machine.
- Benefits: Enables reliable obstacle detection and controlled vehicle response in tight and dynamic warehouse environments. The multi-zone concept allows for gradual deceleration, improving operational smoothness while ensuring a defined safety stop when required. In the presence of stationary obstacles, the system typically detects objects within the outer warning zones at an early stage. This allows the vehicle to decelerate progressively, preventing entry into the final safety (red) zone. As a result, unnecessary emergency stops are avoided, leading to smoother operation, reduced mechanical stress, and minimized wear on braking components.

2.7 Fork Tip Safety bumper

- Purpose: The fork tip safety bumper is a passive mechanical safety protection function for the AGV's front end. It is installed at the front of the fork and is used to directly contact personnel, obstacles, and shelf columns during automatic operation. It triggers the highest priority safety shutdown to prevent crushing, collision, and puncture injuries.
- Working Principle: AGV fork tip safety bumper is a mechanical-contact type safety protection unit mounted at the end of forklift forks. Under normal operating conditions, the elastic bumper keeps in extended position, and the built-in safety switch maintains normally closed status, sending valid enable signal to the AGV safety PLC to permit normal traveling. Once the fork tip collides with personnel, goods or obstacles, external extrusion force pushes the bumper shell to retract along the guide shaft and compress the reset spring. This mechanical displacement immediately actuates the internal safety switch, cuts off the safety circuit signal and outputs emergency stop trigger signal. The AGV system rapidly executes power cutoff and brake deceleration to avoid collision injury and equipment damage. This design adopts fail-safe logic; circuit disconnection or component failure will automatically trigger emergency stop, fully complying with functional safety requirements.
- Benefits: Eliminate the blind area at the tip of the fork, make up for the detection blind spot of the laser radar, and achieve all-round protection for the end of the fork. Fault-safe architecture: In the event of a line break or component failure, an emergency stop is automatically triggered, resulting in higher safety.

2.8 Safety PLC

- Purpose: Central monitoring and control for all safety mechanisms.



- Working Principle: Processes inputs from emergency stops, LiDARs, encoders, and proximity switches; controls speed limits, direction checks, and emergency stops.
- Benefits: ISO 3691-4 and EN ISO 13849-1 compliance, protects personnel and equipment.

2.9 Emergency Stop Switches

- Purpose: Immediate manual shutdown in emergency situations
- Working Principle: Dual independent signals to Safety PLC; disconnects main power contactors for emergency braking.
- Benefits: Redundant safety layer

2.10 3D Lidar Identification

- Purpose: Accurately identify, position and confirm pallet/container carriers for automatic picking.
- Working Principle: 3D laser scans to acquire spatial point cloud data, matches carrier shape, size and posture, realizes precise docking and material pickup.
- Benefits: High positioning accuracy, strong anti-interference, stable recognition, improves automatic loading efficiency.

2.11 Safety Encoder, Rotary Proximity, Load Switch

- Purpose: Monitors speed, direction, and load status.
- Working Principle: Sends signals to PLC; triggers braking or speed limit enforcement under unsafe conditions.
- Benefits: Maintains safe operation under varying loads and turns.

2.12 Acoustic and Light warning Systems

- Purpose: Enhances operational safety by alerting surrounding personnel to vehicle movement, abnormal conditions, and operational status in real time.
- Working Principle: Integrated acoustic alerts, flashing lights, and turn signals indicate task status, faults, warnings, or directional movement.
- Functions:
 - Acoustic alerts for task progress, faults, and warnings
 - Red warning light for forward motion
 - Turn signals for directional changes
 - Automatic alarms triggered by obstacles, communication errors, positioning failures, low battery, or manual operation zones
- Benefits: Ensures personnel safety, situational awareness, and prompt response to operational issues.

2.13 In-Car Interaction

The vehicle provides intuitive onboard human-machine interaction interfaces by touch screen or multi-key button interface. It allows operators to monitor status, perform diagnostics, and execute manual operations when required.

Operator input during automatic mode is implemented in accordance with EN ISO 3691-4 (4.9.2.2). The vehicle stops at defined positions and allows manual actions under



controlled conditions, including hold-to-run operation, no vehicle movement, and a deliberate restart of the automatic sequence.

2.14 Task Triggering and Equipment Interface

Tasks can be triggered using multiple calling devices:

- Button box
- PAD
- Mobile phone
- Computer
- Aerial camera

Purpose: Enables flexible task initiation from different operational positions.

The XQE122 can interact with various fixed and automated systems:

- Automatic doors
- Robotic arms
- Palletizers
- Conveyor lines

Purpose: It allows seamless integration into automated production and logistics systems.

2.15 System Integration

The system supports standard enterprise-level integration with platforms such as WMS (Warehouse Management System), MES (Manufacturing Execution System), and ERP (Enterprise Resource Planning) systems.

Purpose: Enables real-time task dispatching, seamless data exchange, and enhanced operational transparency across intralogistics and production processes.

Cybersecurity: The system is designed with secure communication interfaces and access control mechanisms to ensure reliable and protected data exchange with connected enterprise systems, supporting safe integration within IT infrastructures.

3 Product function definition

The XQE122 performs fully autonomous pallets, boxes and irregular loads transport and storage/retrieval operations inside defined warehouse zones. All movements and handling actions are executed according to tasks received from a fleet management system or triggered manually under controlled conditions. The vehicle integrates navigation, motion control, obstacle detection, pallet alignment, and independent safety supervision to achieve repeatable positioning and safe operation in compliance with ISO 3691-4.



3.1 Navigation system logic description

The XQE122 navigation system enables precise autonomous movement in complex warehouse environments. It combines laser-based 3D SLAM for natural-feature navigation (the default and primary method) , In scenarios with complex and easily lost positioning requirements, QR codes are used for assistance in positioning..The architecture integrates real-time localization, route planning and path following while working in parallel with the obstacle detection and safety systems.

3.1.1 Navigation laser

The default navigation method is natural-environment SLAM. A single navigation laser scanner continuously scans the surroundings and produces a real-time point cloud of the environment. Onboard software matches this live data against a pre-built digital map of the warehouse, allowing the vehicle to always determine its exact position and orientation.



Fig 3.3: Navigation Laser

This method relies solely on existing environmental features (racking, columns, walls, machinery etc.) and requires no additional infrastructure in standard operation. It performs well in typical warehouse layouts with adequate natural variation.

Positioning accuracy remains ± 10 mm under normal conditions (clean floor, stable lighting, no extreme dynamics). The system continuously verifies path adherence and travel direction. If localization confidence drops below threshold, the vehicle slows, attempts recovery using recent data, and stops if needed.

Key functions performed by the navigation system include:

- Continuous real-time localization within the mapped environment
- Map matching between live scan data and the stored map
- Route planning and precise trajectory following
- Verification of forward direction and path adherence

3.2 Person and obstacle detection System

The XQE122 is equipped with a multi-sensor detection system that supports safe and efficient autonomous operation. It provides real-time awareness of obstacles during travel and assists with precise pallet handling during approach, pickup and reversing. The system combines several sensors that operate in parallel and supply processed data to the onboard industrial computer for appropriate speed- and direction-dependent reactions.

Detection zones and responses adapt automatically to the current vehicle speed, travel direction and maneuver type (detailed zone definitions and speed-dependent logic are provided in Chapter 4). When an obstacle is detected in a defined area, the system issues progressive warnings, reduces speed or applies controlled braking as required. These



detection functions improve operational smoothness and situational awareness but are separate from the independent, certified safety supervision layer (see 3.5).

3.2.1 Blind-spot laser Lidar

The Blind-spot laser Lidar function refers to an intelligent function that utilizes a three-dimensional lidar (LIDAR) to scan the surrounding spatial environment in real-time, perceive the three-dimensional position and shape of obstacles, and thereby achieve dynamic obstacle avoidance. 3D lidars can acquire height information, enabling the identification of suspended obstacles, low obstacles, and complex terrains such as protrusions at the bottom of shelves, hanging objects, slopes, etc., enhancing the comprehensiveness and safety of obstacle avoidance.

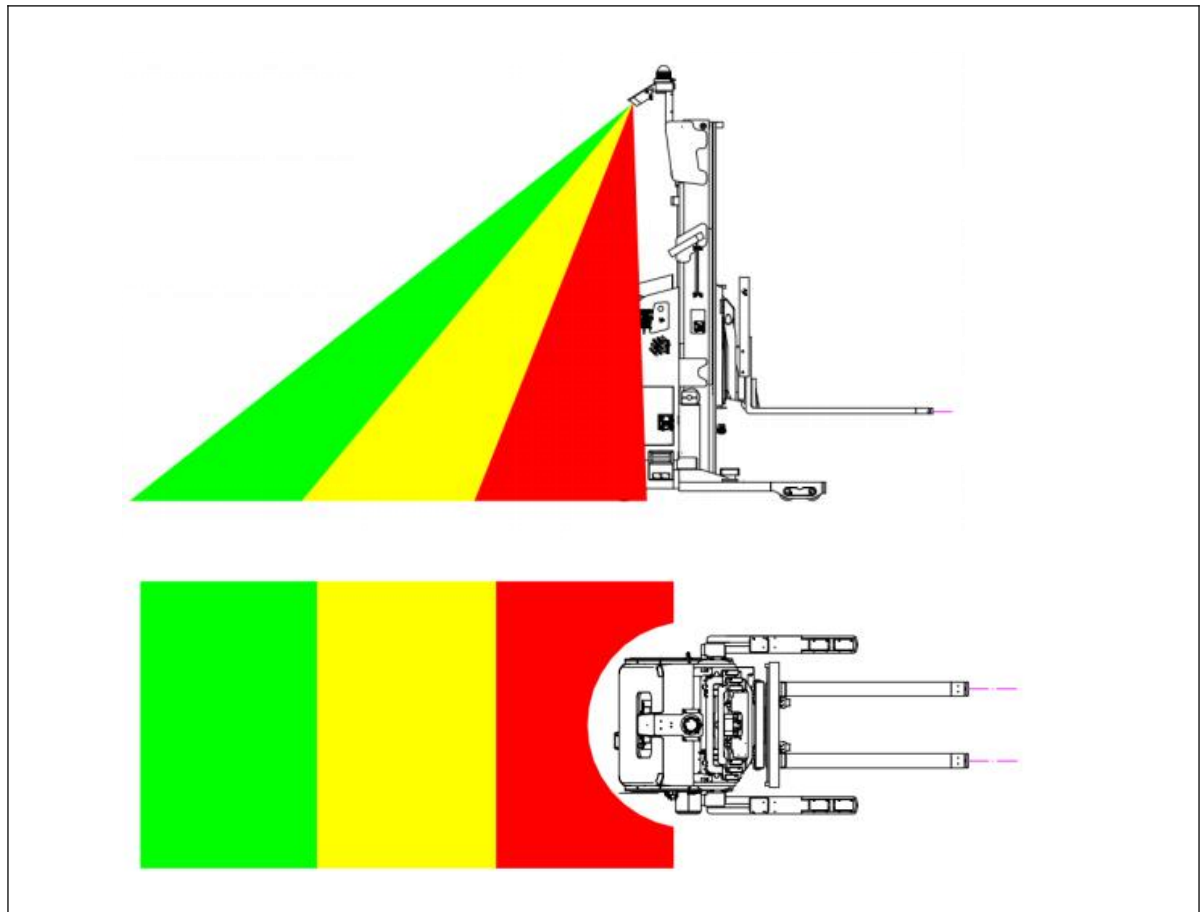


Fig 3.4: Blind-spot laser Lidar

The system constructs a real-time point cloud model that enables the system to identify objects at different distances and initiate the following operational responses:

- When an obstacle enters the warning zone (yellow), an audible alert and visual indication are activated, and the vehicle significantly reduces speed.
- When an obstacle enters the critical zone (orange), the vehicle comes to a full stop using torque braking commanded through the F1-511 traction controller.



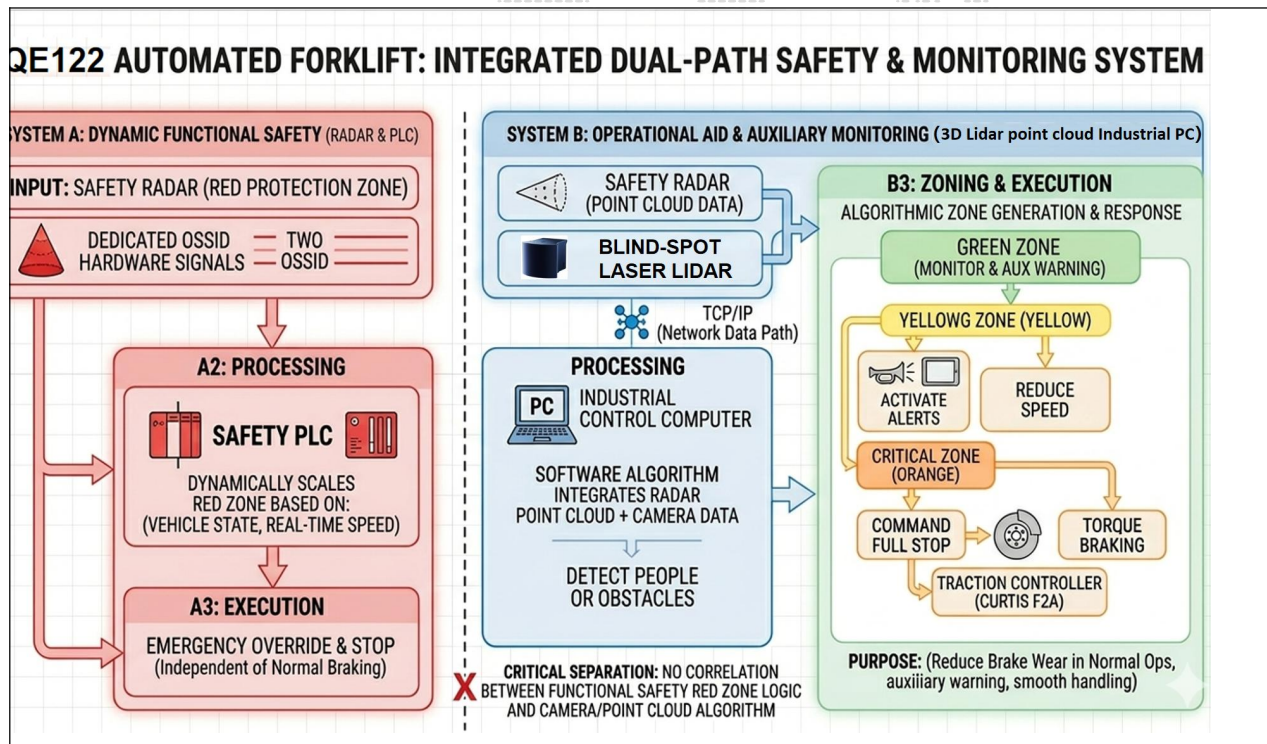


Fig 3.5: Logical Diagram

3.2.2 360° PLD LiDAR

Three LiDARs (Located on the front of the vehicle and on the supporting legs on both sides) provide continuous 360° ground-level coverage around the vehicle perimeter. These sensors detect low-level obstacles such as pallets on the floor, protruding legs, pedestrians' feet, small items or other ground hazards that could interfere with safe movement.

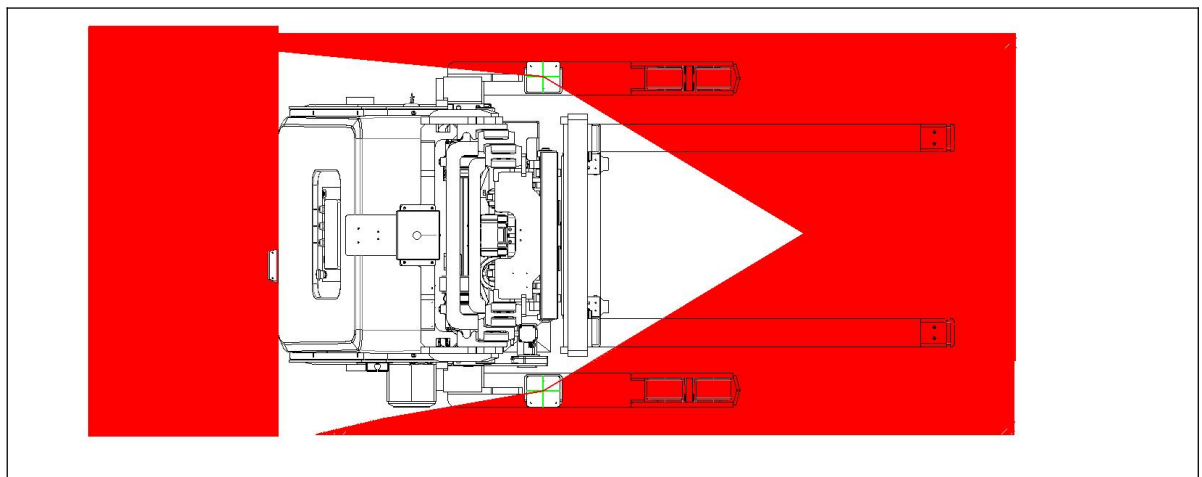


Fig 3.6: Bottom safety LiDAR

The LiDARs generate real-time point-cloud data that is processed to define multiple concentric protection zones.

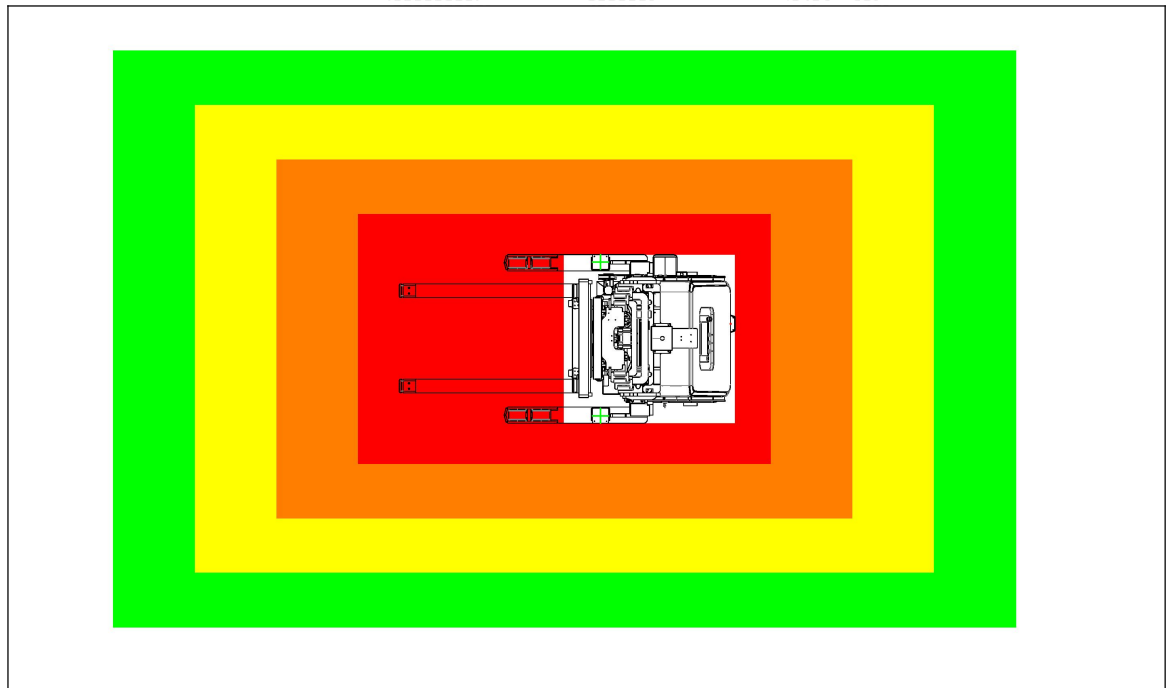


Fig 3.7: LiDAR Zones

These zones adjust automatically based on the vehicle's speed, direction of travel (forward, reverse or turning) and maneuver type. The response logic is progressive:

- Warning area (green): Audible and visual alert issued; vehicle applies moderate deceleration to allow time for the obstacle to clear or for the AGV to slow safely.
- Deceleration area (yellow): Stronger controlled deceleration activated via torque braking through the F1-511 traction controller.
- Parking/stop area (orange): Vehicle comes to a full controlled stop using torque braking.
- Red (safety zone): This innermost zone is monitored by the independent Safety PLC (Category 3 PL d rated LiDARs according to EN ISO 13849-1 feeding into PL e/d safety chain). Entry into the red zone triggers immediate opening of dual contactors (KM1 and KM2), cutting power to both the F1-511 traction controller and the F1-T4 steering controller. The spring-applied electromagnetic brake (EB) engages for a fail-safe stop. Resumption of motion requires obstacle clearance, manual reset, and a 3-second delay with audible and visual warnings.



3.3 Safety control system

XQE122 is equipped with an independent safety control system designed to meet the requirements of ISO 3691-4. This system operates separately from the navigation and task execution software and has the highest authority to interrupt motion and force a fail-safe stop when necessary.

3.3.1 Safety PLC

The safety PLC is the central monitoring unit of the safety system. It continuously supervises critical vehicle parameters and enforces safe operation according to ISO 3691-4 performance level requirements.

Its main functions are:

- Safety speed monitoring detects overspeed, stall, wrong direction, or speed limit violations (forward/reverse, loaded/unloaded) using the safety encoder.
- Safety direction monitoring recognizes left/right turn direction via rotary proximity switches and switches LiDAR protection zones automatically.
- Safety LiDAR monitoring: independently supervises the bottom 360° safety LiDAR protection fields.
- Safety emergency stop: processes signals from all emergency stop switches.
- Additional safety signal monitoring: supervises load switch status, bumper activation, contactor feedback, and other I/O signals.



Fig 3.8: Safety PLC

3.3.2 Emergency stop switch

The XQE122 is equipped with four emergency stop switches to ensure rapid access and compliance with ISO 3691-4 and ISO 13850:

- Two side-mounted emergency stop switches (one on each side of the vehicle, positioned for easy reach from the ground or during maintenance).
- An emergency stop switch (located at the front display operation interface for ease of operation and maintenance)



- One central/main emergency stop switch (The red button is located on the right side of the vehicle, clearly visible and easy for personnel to operate.).

Each switch provides two independent normally closed (NC) signals connected directly to the safety PLC. The PLC continuously monitors the status of all three switches in real time.

When any one of the emergency stop switches is pressed (whether in automatic or manual mode), the safety PLC immediately detects the signal disconnection. It then:

- Opens the two main contactors (KM1 and KM2).
- Cuts power to the F1-511 traction controller and F1-T4 steering controller.
- Engages the spring-applied electromagnetic brake (EB) for a fail-safe emergency stop.



Fig 3.9: Emergency Stop Switch

The system uses closed-loop verification: the PLC monitors the normally closed auxiliary contacts of the main contactors to confirm that power has been successfully removed.

To resume operation after an emergency stop:

- The obstacle or fault condition must be cleared.
- The pressed emergency stop switch(es) must be released (twisted and pulled back to the normal position).
- The operator presses the reset button.
- After a mandatory 3-second delay accompanied by audible and visual warnings, the vehicle is permitted to r
- estart and continue the task.



4 Operating Conditions

This chapter defines the permissible environmental conditions, ground/floor requirements, personnel safety rules, and typical usage scenarios for the XQE122 automated pallet truck. Compliance ensures stable navigation (± 10 mm accuracy), reliable performance, and safe operation in accordance with ISO 3691-4.

4.1 Operating requirements

4.1.1 Power requirements

- Nominal voltage: 230 V AC, single-phase, 50 Hz
- Permissible fluctuation: $\pm 10\%$
- Protective grounding (earth) required in the power distribution system for electrical safety
- No specific protection for connection plates mentioned; standard industrial connectors are used (ensure proper strain relief and IP rating per site conditions)

4.1.2 Equipment operating environment requirements

The XQE122 is designed for indoor industrial environments with the following limits:

- Operating temperature: $-10\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$
- Charging temperature: $0\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$ (battery type dependent; consult manual for exact limits)
- Relative humidity: up to 90 % RH, non-condensing
- Avoid direct sunlight on laser navigation sensors
- Avoid large reflective surfaces (glass, mirrors, polished metal) or pure black/absorbing areas in the laser scan field; if unavoidable, frost glass or shield surfaces
- No corrosive gases, explosive atmospheres, or harmful liquids
- Low dust and metal particle concentration
- No continuous vibration or excessive mechanical shock

4.1.3 Floor requirements

The driving surface must be smooth, clean, and stable to support reliable laser SLAM navigation and mechanical performance. Requirements follow general AGV guidelines (N 13036-7, VDI 2198, ISO 3691-4 expectations):

Surface Undulation:

Undulation is defined as the height difference between the highest and lowest points within a reference area.

- Maximum allowable undulation within 1 m^2 : $\leq 3\text{ mm}$
- The surface must be:
 - Clean
 - Free of loose particles and

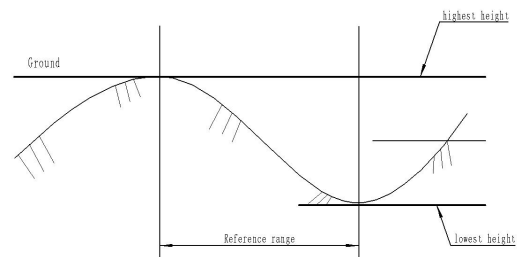


Figure Road Surface Roughness Requirements



- debris
- Non-slip

Figure 5.1. Road surface smoothness requirements

Excessive undulation may cause vibration, reduced navigation accuracy, and increased wear on the wheel and drive system. Maintaining a smooth surface ensures stable motion and precise positioning.

Road Surface Slope

- Road surface slope (H/L) is defined as the ratio between height difference (H) and route length (L) over a minimum distance of 100 mm.
- Maximum allowable slope for travel: 8-10%
- Maximum allowable slope for precise parking points (H/L): $\leq 1\%$

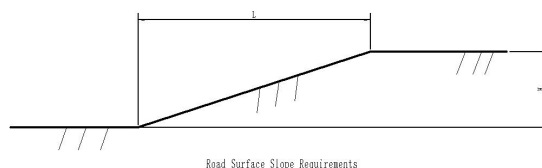


Fig 5.2: picture Road slope requirements

Step height:

Maximum allowable step height: ≤ 5 mm

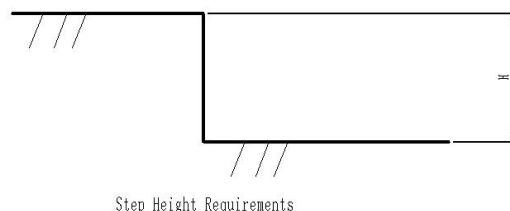


Fig 5.3: Step height requirements

Groove Width:

Groove width refers to the maximum width of gaps or channels in the floor surface.

- Maximum allowable groove width: ≤ 8 mm
- If groove width exceeds this value, step height requirements apply

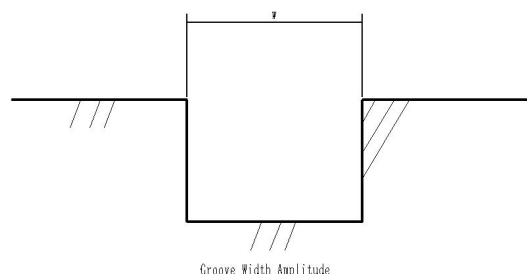


Fig 5.4: Groove width

Load-bearing requirements:



- Static load calculation:

$$\text{Static load} = \frac{\text{Vehicle weight} + \text{Load}}{\text{Wheel system projected area}} \quad (\text{kg/m}^2)$$

- Dynamic load capacity must be $\geq 150\%$ of static load
- Under maximum load, uneven ground settlement must be $< 1/1000$

Floor requirements:

- Emery or hardened cement floors must be:
 - Wear-resistant
 - Dust-resistant
 - Anti-static
- Epoxy floors must receive anti-static treatment
- Waxing is prohibited
- Static friction coefficient (floor vs. PU wheels): ≥ 0.55
- Floor hardness: C30 or higher (≥ 30 MPa)

4.1.4 Shelf Assembly Requirements

Compliant with EN 15512, EN 15620, EN 15635 and CE marking (per CPR 305/2011/EU & Machinery Directive).

- The floor shall have a flatness tolerance ≤ 3 mm/m and sufficient load-bearing capacity.
- Uprights must be installed with verticality $\leq 1/500$ of height, base plates fully seated, and properly anchored.
- Beams shall be level within ≤ 2 mm/m, securely locked with no missing fasteners, and deflection shall not exceed $L/200$ under rated load.
- Assembly tolerances include frame spacing ± 5 mm, beam height ± 3 mm, and system squareness ≤ 10 mm per 10 m.
- All bracing and stabilizers must be fully installed; floor anchoring is mandatory for heights above 3 m, and no unauthorized modifications are allowed.
- Installation shall be performed by certified personnel, with as-built drawings, tolerance inspection reports, and load capacity labels provided.

4.1.5 Personnel safety requirements

To ensure safe coexistence with personnel, the following rules apply:

- Personnel must actively avoid the AGV path and never intentionally obstruct or collide with it
- In narrow aisles (< 3.5 m), the AGV reduces speed and detection range; no sudden entry of people or obstacles permitted
- Keep routes clean and free of foreign objects (screws, hoses, cables, etc.)
- Do not modify drive mechanisms, control cabinets, or sensors
- Only trained and authorized operators may interact with the system
- No unauthorized electrical or software modifications
- Observe mechanical hazard warnings near moving parts



- Non-authorized persons must not access control panels during operation
- In emergencies, press any emergency stop switch and await qualified personnel
- Do not block LiDARs, cameras, or sensors
- The system is not waterproof or explosion-proof
- Perform visual inspection of mechanical/electrical components before startup
- Do not sit, strike, climb, lean on, or ride the AGV or load
- Cleaning permitted only when AGV is stopped and powered off

Minimum safety distance: Personnel should maintain at least 0.5 m clearance from the AGV envelope during motion (increased in narrow aisles per site risk assessment).

4.1.6 Usage scenario requirements

The XQE122 vehicle is a compact indoor extendable stacker-type automatic handling robot, suitable for high-rise stacking (with a lifting height of up to 5.5 meters) and transportation operations in narrow passages.

- Standard warehouse scenario;
- Deep alley warehouse scenario;
- Rack-to-rack connection scenario;
- Automatic door docking scenario;

Applicable to indoor warehousing scenarios including shelf warehouses and stacking warehouses. It supports high-level stacking of material frames, pallet putaway, automated warehouse docking and buffer area sorting.

Equipped with automatic travelling, forklift unloading, shelf loading & unloading, targeted shelf stacking and obstacle avoidance functions. A single unit fulfills multiple workflows such as goods receiving, outbound delivery, overnight pre-sorting, production line feeding and automated warehouse connection. It serves as an optimal solution for intensive storage and digital warehouse upgrading.

4.2 Aisle width requirements for each scenario

Aisle widths depend on load configuration, turning radius, speed, and single/dual-lane operation. Minimum values (based on vehicle kinematics and safety clearance) are:

Table 5.1: Aisle width Requirement

Load Carrier	Operation Type	Aisle Type	Minimum Aisle Width	Remarks
Euro Pallet / Cage / Bin	Straight Travel	Single Aisle (Through Traffic Only)	1600 mm	Speed $\leq$ 0.3 m/s
Euro Pallet / Cage / Bin	Straight Travel	Dual Aisle (Through Traffic Only)	3360 mm	
Euro Pallet / Cage / Bin	Turn	Single Aisle (90° Turn)	2450 mm	
Euro Pallet / Load Carrier $\leq$ Euro Pallet	Pick-up / Drop-off	Single Aisle (Single-side Automated Pick-up/Drop-off)	2900 mm	



Load Carrier	Operation Type	Aisle Type	Minimum Aisle Width	Remarks
Size		Single Aisle (Dual-side Automated Pick-up/Drop-off)	2900 mm	Two main aisles between racks on both sides
		Single Aisle (Dual-side Automated Pick-up/Drop-off)	3520 mm	One main aisle between racks on both sides
Cage / Bin, L×W: 1200×1200 mm	Pick-up / Drop-off	Single Aisle (Single-side Automated Pick-up/Drop-off)	3000 mm	
		Single Aisle (Dual-side Automated Pick-up/Drop-off)	3000 mm	Two main aisles between racks on both sides
		Single Aisle (Dual-side Automated Pick-up/Drop-off)	3520 mm	One main aisle between racks on both sides
Euro Pallet / Cage / Bin	Straight / Reverse	Single Aisle (High-Density Stacking)	1640 mm	Speed ≤ 0.3 m/s

NOTE

- 1. European standard pallet dimensions: L*W*H: 800mm*1200mm*144mm, L*W*H: 1000mm*1200mm*144mm;
- 2. Material cages or boxes, dimensions ≤ 1200*1200mm;
- 3. The maximum dimensions of all above carriers shall not exceed L*W: 1200*1200mm. Floor/ground markings must comply with ISO 3691-4 Annex A (Table A.1/A.2) for zone definition and personnel awareness (restricted, confined, hazard zones).



5 Wi-Fi communication

The XQE122 uses industrial-grade Wi-Fi for reliable two-way communication with the fleet management system (dispatch server). This connection enables real-time task management, status reporting, and safe operation across the warehouse;

5.1 Wi-Fi Communication Module

The Wi-Fi module is the primary interface for data exchange between the AGV and the dispatch system. Its key functions include:

- Receiving structured task commands (pickup location, drop-off location, action type) from the fleet manager
- Transmitting real-time vehicle data: current position, operational status, battery level, alarms/warnings, and task progress
- Supporting dynamic rescheduling, fault reporting, and emergency interventions

The module operates on a stable industrial wireless network (2.4 GHz and 5 GHz bands) with high throughput and low latency. It ensures continuous connectivity within the defined operating area, allowing efficient task execution even in environments with moving obstacles or multiple vehicles.



Fig 6.1: Industrial Grade Wi-Fi Module

5.2 WiFi Requirements

project	require	remark
signal strength	≤ -50 dBm	The minimum requirement for a stable connection
delay	≤ 50 ms	Ensure that the scheduling system responds in real time.
frequency band	2.4G / 5G dual-band	The 2.4G has strong penetration capability and is more stable; the 5G has a higher speed.
Wandering Switch	≤ 100 ms	The vehicle continuously connects when crossing the AP.
coverage area	seamless	It is necessary to cover the entire driving route and charging stations.



5.3 Wireless communication for automatic charging

The XQE122 supports automatic charging stations with wireless docking. The interaction follows a defined sequence coordinated by the fleet management server:

1. The server assigns a charging task and directs the AGV to a designated charging station.
2. Upon arrival, the AGV reports its position to the server.
3. The server instructs the charging station to initiate the charging process.
4. The charging station extends its telescopic charging arm, docks with the AGV's charging interface, and begins charging.
5. Continuous communication with FMS and Charger in order to monitor the status of charging.
6. When charging is complete (or interrupted), the server sends a stop-charging command.
7. The charging station retracts the telescopic arm to its fully retracted position.
8. The station confirms full retraction to the server (via feedback signal).
9. The server authorizes the AGV to depart safely.

6 Standard pallet information

The XQE122 is designed to handle standard pallets, boxes and irregular loads.

EPAL Euro pallet: Standard dimensions (as per EPAL specification):

- Length: 1200 mm
- Width: 800 mm
- Height: 144 mm
- Deck boards: 22 mm thick, 145 mm wide (top), 100 mm wide (bottom)
- Stringers: 145 mm high, 100 mm wide
- Blocks: 145 mm × 100 mm × 78 mm (corner), 145 mm × 145 mm × 78 mm (center)
- Fork entry: Four-way access, with clearance optimized for AGV forks.
- Refer to the illustration below for detailed EPAL pallet dimensions and structure.
- (Fig 8.2: EPAL Pallet Dimensions)



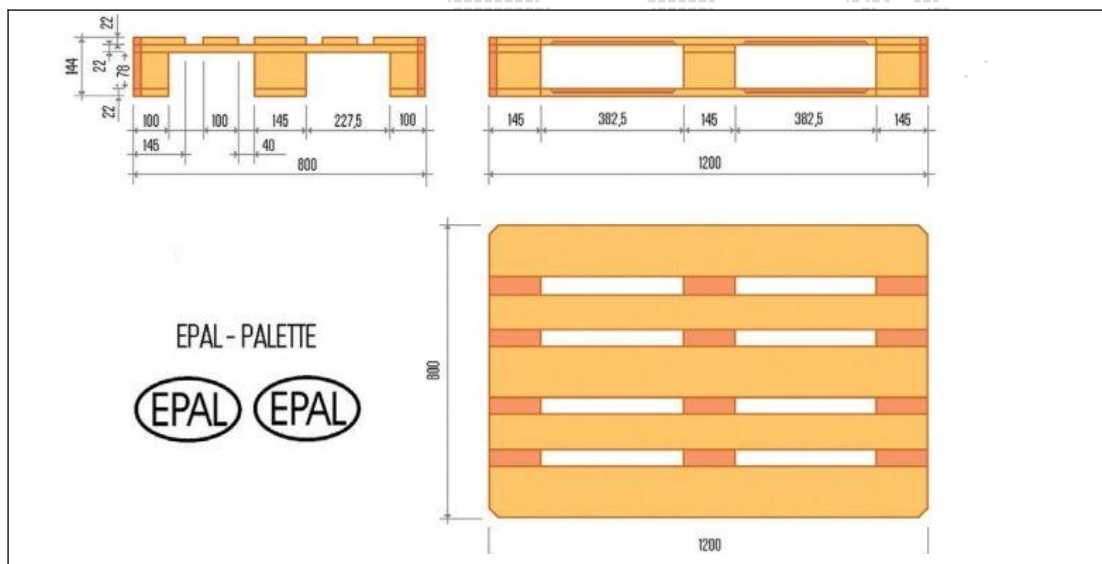


Fig 8.2: Pallet Dimensions

Note:

- 1、 The size of the material cage and the material box is determined according to the actual requirements of the customer;
- 2、 the width of the carrier should not exceed the width of the vehicle body. If it needs to be wider, it is necessary to communicate and confirm with the relevant scheme designers.

7 Shelves and goods requirements

7.1 Shelf requirements

European standard pallet dimensions: L*W*H: 800mm*1200mm*144mm;

Length of bridge	Shelf depth	Number of trays	Tray spacing
1800 mm	1100mm	2	≈75 mm
2700 mm	1100mm	3	≈75 mm
3600mm	1100mm	4	≈80mm

European standard pallet dimensions: L*W*H: 1000mm*1200mm*144mm;

Length of bridge	Shelf depth	Number of trays	Tray spacing
2700 mm	1100mm	2	≈75~100 mm

7.2 cargo requirement

Dimensions for reserved space in goods layout:

7.1 Left-right clearance: The edge of the goods should be ≥ 50 mm away from the edge of the pallet;

7.2 Front-back clearance: The edge of the goods should be ≥ 50 mm away from the edge of the pallet;

7.3 Suspended clearance: The goods should not exceed the range of the pallet;



- 7.4 Center of gravity height: When fully loaded, the center of gravity height of the goods should be $\leq 900\text{mm}$ (from the fork face);
- 7.5 Stack alignment: The deviation of the edges of the goods in the upper and lower layers should be $\leq 10\text{mm}$
- 7.6 Safe spacing of materials in the rack from left to right: $\geq 75\text{mm}$ (net space), recommended value $\geq 100\text{mm}$;
- 7.7 Spacing between adjacent racks or wall-mounted racks: $\geq 200\text{mm}$ (when the vehicle does not exceed the rack);
- 7.8 Spacing of materials in the rack from top to bottom: $\geq 150\text{mm}$ (net space);

